baboon will feel confused and indecisive. 'Yes…No…Yes…No… Damn! I don't know what to do!'

In order to transmit genes to the next generation, it is not enough to solve survival problems. Animals also need to solve reproduction problems too, and this depends on calculating probabilities. Natural selection evolved passion and disgust as quick algorithms for evaluating reproduction odds. Beauty means 'good chances for having successful offspring'. When a woman sees a man and thinks, 'Wow! He is gorgeous!' and when a peahen sees a peacock and thinks, 'Jesus! What a tail!' they are doing something similar to the automatic vending machine. As light reflected from the male's body hits their retinas, extremely powerful algorithms honed by millions of years of evolution kick in. Within a few milliseconds the algorithms convert tiny cues in the male's external appearance into reproduction probabilities, and reach the conclusion: 'In all likelihood, this is a very healthy and fertile male, with excellent genes. If I mate with him, my offspring are also likely to enjoy good health and excellent genes.' Of course, this conclusion is not spelled out in words or numbers, but in the fiery itch of sexual attraction. Peahens, and most women, don't make such calculations with pen and paper. They just feel them.

Even Nobel laureates in economics make only a tiny fraction of their decisions using pen, paper and calculator; 99 per cent of our decisions – including the most important life choices concerning spouses, careers and habitats – are made by the highly refined algorithms we call sensations, emotions and desires.[18]

Because these algorithms control the lives of all mammals and birds (and probably some reptiles and even fish), when humans, baboons and pigs feel fear, similar neurological processes take place in similar brain areas. It is therefore likely that frightened humans, frightened baboons and frightened pigs have similar experiences.[19]

There are differences too, of course. Pigs don't seem to experience the extremes of compassion and cruelty that characterise *Homo sapiens*, nor the sense of wonder that overwhelms a human gazing up at the infinitude of a starry sky. It